

Coming to the University of Virginia, we didn't think an unfamiliar event app would end up being such a big part of our college life, but DoorList quickly changed that. It didn't take long to see how much people rely on it: from academic and club meetings to Greek life events, rush activities, bar tabs, fundraisers, and even just casual hangouts. Whether you're hosting or attending, DoorList makes event organization much easier.

DoorList is used by college students, organizations, and event hosts to manage and check people into events easily. Traditionally, organizations or event hosts have relied on physical check-in lists or wrist bands. DoorList replaces this traditional approach with a digital QR code used for entry. It is deeply rooted in UVA culture, having been created by a UVA alumnus who recognized the need to streamline attendance tracking for both hosts and attendees. Since its launch at UVA, the app has expanded to other universities such as University of North Carolina and the University of Texas.

With DoorList being such an integral part of UVA's campus culture, we saw value in analyzing its impact, especially since it is a tool we personally rely on in our own organizations. We believe our analysis can help organizations better understand their hosting patterns and improve how they plan events and manage resources, while also providing DoorList with insights to enhance its platform. Through our findings and suggestions, we aim to benefit not only UVA students but also those at other universities as DoorList continues to grow.

As a means of jumpstarting our project, we reached out to some fraternities to request access to their data. We soon realized there would be numerous obstacles to obtaining it, and any data we did receive would likely be incomplete. In addition, our sample size would be quite small and inaccurate as it would only include some fraternities. This approach would not give us access to individual hosts' data or other types of hosts, like academic clubs or sororities.

Once we realized that acquiring data from fraternities was not a viable option, we scheduled a call with DoorList. We connected with their team and arranged a Zoom meeting with UVA alumnus Mitchell Whalen, a growth associate. After discussing our capstone goals, we were granted access to DoorList's sensitive data upon signing an NDA to ensure the privacy of attendees, hosts, and organizations. Accessing the data directly from DoorList was ideal, as it ensured accuracy and reliability. We met with Mitchell multiple times to review the dataset, gaining a clear understanding of each variable and how to use the information effectively.

The data we received is at the event level, meaning each row represents a single event rather than an individual attendee, which allowed us to focus on how events differ from one another. It covers a wide range of UVA DoorList events, including big parties, meetings, date functions, small gatherings, and other social events, spanning from August 5th, 2024, to September 8th, 2025. Although the data was in JSON format, we decided to use the information available and turn it into something workable for our analysis.

The data contained 1,213 rows of UVA events. There were 117 columns in the original dataset, including many predictor variables and our response variable, `thruDoor`, which lists the IDs of people who attended an event via DoorList's scanning feature. After discussions with DoorList representatives,

we concluded that including all 116 predictor variables in our model would not be practical. Many predictors were irrelevant to our analysis, some contained mostly missing values, others were obsolete and no longer in use, and, most importantly, having too many predictors would hinder our model's performance. Therefore, we decided to filter the dataset to only include 13 of those original 116 predictors. Additionally, DoorList representatives gave us a list of organization names that were associated with test cases. We filtered the dataset by removing these test cases and any cancelled events.

Next, in order to better format the JSON data for our analysis, we had to parse through fields and make additional columns with these extracted values. First, we parsed through the ``thruDoor`` column, our response variable, by counting the number of commas separating the different attendee IDs and adding one to accurately represent the total number of attendees. If there was an empty list of IDs, the attendee count for that event was set to zero. After parsing ``thruDoor``, we created a new column, ``attendee_count``, which counts the number of attendees at each event. We followed the same approach for other columns like ``approvedID``, which counts the number of approved IDs for an event and totals them in a column named ``numApprovedID``. For other columns that we needed to parse through, such as ``requestedPlusOnes``, ``going``, ``maybe``, ``notGoing``, ``allowedOrgs``, and ``orgPlusOnes``, we followed similar procedures to count the IDs and made columns such as ``totRequestedPlusOnes``, ``RSVPyes``, ``RSVPmaybe``, ``RSVPnotGoing``, ``numAllowedOrgs``, and ``orgPlusOnes_num`` respectively. Finally, we changed the time variable from UTC format to Eastern time, and extracted both the hour and day of the week from each timestamp to make the columns ``event_hour`` and ``event_day``. ``event_hour`` is represented by a number from 0 to 23, and ``event_day`` is represented by the day of the week. With these new variables, we were able to make our time-based analyses more interpretable and uncover patterns in event activity throughout the day and week.

During our initial exploratory data analysis, we observed that many events in the dataset had attendee counts of zero. To address this, we decided to focus on a subset of events: those with at least one invited organization, regardless of attendance size, and those without invited organizations only if they had ten or more attendees. We made this choice because many events appeared to be "joke" events created by users that never actually took place. However, we still included events with zero attendees if at least one organization had been invited, since some legitimate events are listed on DoorList but have no scans, which is a pattern we have observed from experience. This approach allowed us to retain valid but unscanned events while filtering out those that were likely invalid. After filtering, we were left with 635 events in our dataset.

Based on our understanding of UVA events and DoorList, we decided to add a new variable, ``event_type``, to classify each event. Since our filtered dataset included 635 events and our team consists of four of us, we decided to divide the events among ourselves and individually classify each type. We came up with eight categories to classify each event: big party, mixer (get together with 2-4 frats/sororities), me + 3 (fraternity/sorority party where each person in the organization gets to invite three guests), rush event, meeting, date function, small event, and other. We relied on our judgement and experience as UVA students to assign each event to the most appropriate category. To classify events in our model, we created a data dictionary that maps specific keywords from event names to their

corresponding categories. For example, for date functions some key words are “dfunc”, “date function”, “df”, etc.

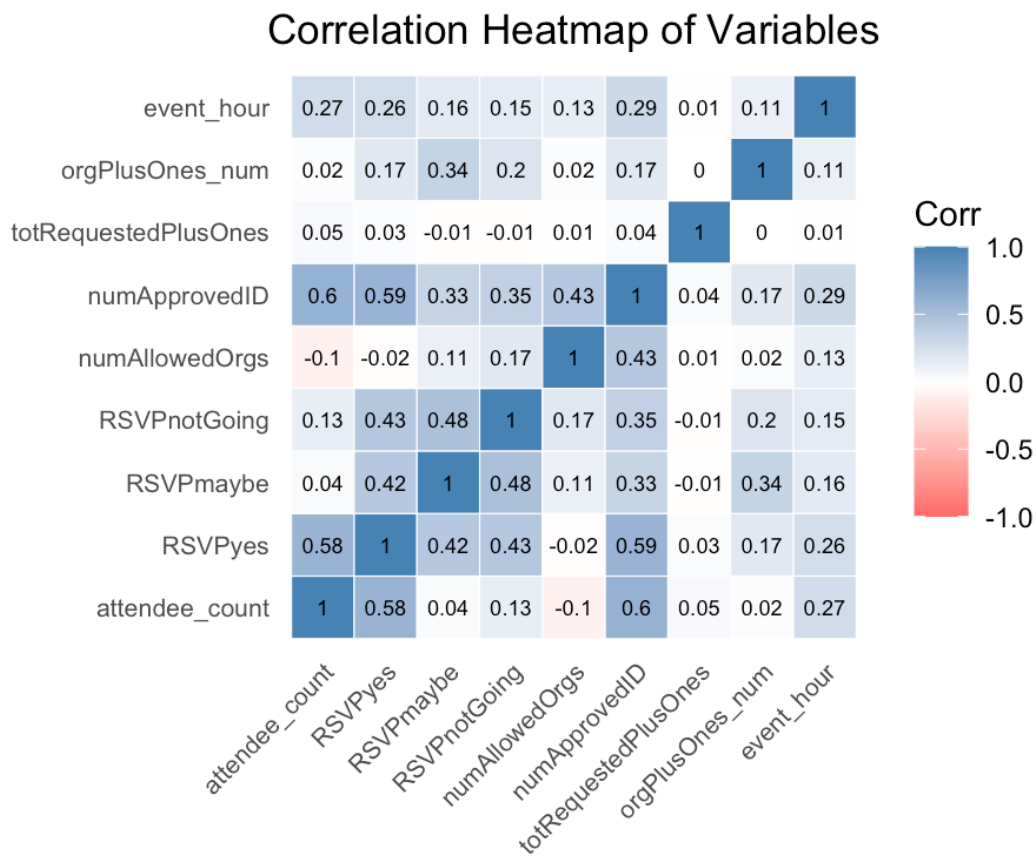
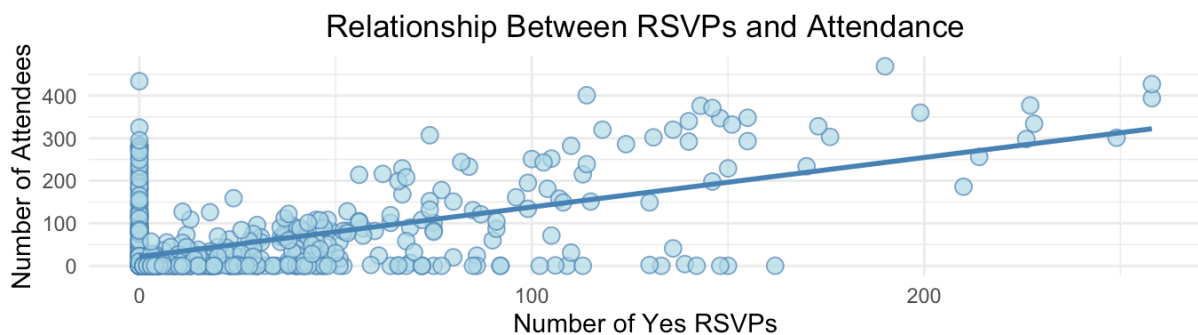
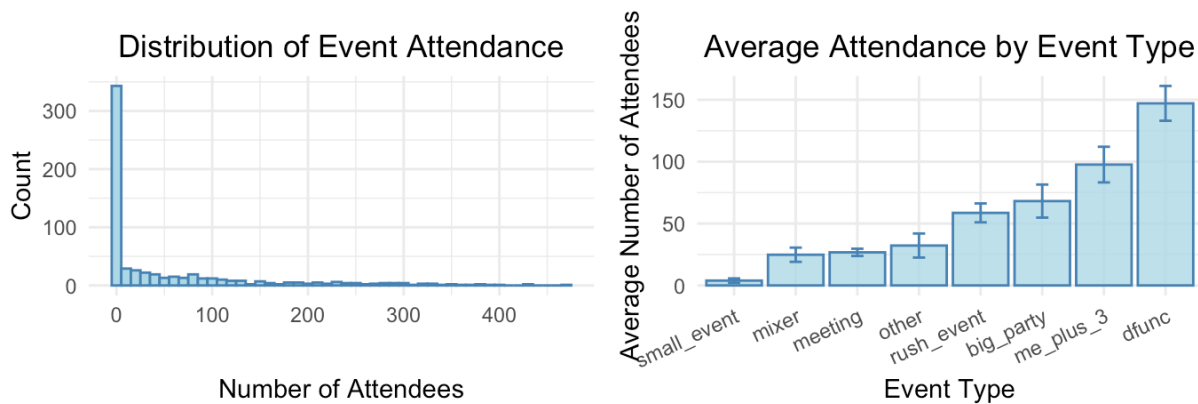
Since our data is structured at the event level, the main research question we chose to study is: How many people actually attend an organization’s events? We decided on this question because it closely aligns with DoorList’s purpose. The app is centered around tracking attendance and aiding hosts in organizing who comes to their events. Rather than measuring how “successful” an event is, our goal is to predict how many people attend so hosts can plan more effectively. By examining factors such as event type, time and day, number of invited organizations, RSVP responses, among others, we can hopefully identify characteristics closely associated with event turnout. Understanding these patterns can help organizations plan their events more effectively and make smarter decisions about scheduling, capacity, and preparation.

While the dataset we received from DoorList was substantial, there were still a few things we wish we could have included. The main limitation was that our data was at the event level instead of the individual level. Having individual-level data would have allowed us to explore student engagement on Grounds in a more specific way. For example, we could have potentially found patterns like whether friends or people in the same organizations tend to go to similar events, if students in certain groups prefer particular event types over others, and how people’s attendance habits change over time. It would have been interesting to study and include how individual and group dynamics contribute to event attendance.

We also wish we could have had access to more than one year of data. Multiple years of data would have provided more information about how events and attendance have changed over time and whether any long-term patterns exist. It would have been helpful to look at recurring events to see if attendance stayed consistent or shifted from year to year, or if certain event types became more or less common over time. With additional years of data, we also could have examined whether hosting activity varies by semester to see if certain parts of the year consistently have higher event activity and attendance than others.

In addition, there were a few other pieces of information that would have been beneficial for our analysis. For example, weather data could have helped explain variation in turnout, since factors like rain, cold temperatures, or warm, sunny days can affect whether people choose to attend an event. It also would have been useful to know if there were other non-DoorList events occurring at the same time, since those could draw attendance away from DoorList events. Finally, how events were promoted, such as whether they were advertised more publicly across Grounds or mainly within smaller groups or organizations, would have been helpful. This additional information could have given us a clearer understanding of what factors drive attendance and how external conditions might influence attendance patterns.

After cleaning and preparing the data, we explored some general patterns in attendance and related variables before entering the modeling phase.



Demonstrated in the top-left plot in Figure 1, the distribution of attendance was strongly right-skewed, with most events drawing only a handful of people and only a small subset of events with high turnout. Attendance also varies by event type, with small events, mixers, and meetings averaging fewer attendees, while date functions and me plus three events attracting larger crowds, which is shown in the “Average Attendance by Event Type” bar chart.

The scatterplot between RSVPs and attendance shows a clear positive relationship, in which events with more “Yes” RSVPs tend to have higher attendance. The cluster of points along the x-axis at zero likely represent events where organizations did not scan attendees, resulting in an attendance of zero despite having RSVPs. Conversely, the points along the y-axis at zero likely reflect events with no RSVPs where attendees still showed up to the event.

Figure 2 displays the distribution of attendance across days of the week. Thursday and Friday show the greatest variability in attendance, with the largest IQRs, while Saturday has a smaller IQR but several high outliers. Sunday also shows moderately wide variation. Based on these patterns and the context of UVA event culture, we grouped days into two categories: weekdays and weekends, defining weekends as Thursday at 12 AM through Sunday at 2 AM, with all remaining days classified as weekdays.

Finally, the correlation heatmap in Figure 3 highlights which numeric variables are most closely related to attendance. The two strongest relationships were both positive. Specifically, ``attendee_count`` was moderately correlated with ``numApprovedID`` (0.6) and ``RSVPyes`` (0.58), meaning that events with more approved IDs on the guest list and more “Yes” RSVPs tended to have higher turnout. Attendance also showed a weaker positive correlation with ``event_hour`` (0.27), indicating that later events may draw slightly more people. All other correlations with attendance were very small, indicating that variables like requested plus-ones, the number of invited organizations, and maybe/not going RSVPs do not appear to have much of a linear relationship with turnout. Overall, this provided us a better idea of which variables seemed potentially meaningful as we moved into the modeling stages.

Finally, Figure 3 presents a correlation heatmap showing which numeric variables are most related to attendance. The strongest positive correlations were with `numApprovedID` (0.6) and `RSVPyes` (0.58), indicating that events with more approved IDs and more “Yes” RSVPs tend to have higher turnout. Attendance was also weakly correlated with `event_hour` (0.27), suggesting slightly higher turnout for later events. All other correlations were minimal, implying that they have little linear relationship with attendance. These insights helped guide which variables to prioritize in our modeling.

Regarding our analytical approach, we utilized multiple linear regression with stepwise model selection based on the Akaike Information Criterion (AIC) in R. This method provided us a systematic way to identify the model that best explains variation in the response variable while maintaining a good balance between fit and complexity. Specifically, our analysis began by fitting a full model that included all theoretically relevant predictors and possible interaction terms. The ``stepAIC()`` function then iteratively added and removed predictors, calculating the AIC at each step and selecting the model with the lowest AIC value. Including the interaction term between ``numAllowedOrgs`` and ``numApprovedID`` allowed the model to capture how the combined effect of the number of organizations invited and the total number of people invited influences attendance, which may improve model accuracy.

Multiple linear regression was selected because it offers a clear framework for quantifying relationships between multiple explanatory variables and a continuous dependent variable. It allows for direct interpretation of coefficients, which aligns with our goal of making the results intuitive and easy to communicate with our audience. It allows us to understand which predictors contribute most strongly to the outcome and in what direction. The stepwise AIC approach is particularly appropriate here because it balances model complexity with explanatory power, automatically penalizing models that include unnecessary predictors. This method ensures that our final model is both statistically robust and interpretable.

Before finalizing our model by using multiple linear regression and stepwise selection, we also considered ridge regression. Given our limited number of observations, we thought this would be a logical method to improve our prediction model. Ridge regression optimizes the model by “shrinking” the beta coefficients so they are all closer to zero. We decided to choose the regularization parameter automatically using cross-validation with the `cv.glmnet()` function. Although we expected ridge regression to enhance our model, its R-squared value was lower than that of our multiple linear regression model. We also preferred multiple linear regression for its greater interpretability, since ridge regression shrinks coefficients. Additionally, ridge regression is most beneficial when there is high multicollinearity between variables, which was not a concern in our dataset.

Although multiple linear regression has many strengths, it also has some notable limitations. The method assumes a linear relationship between predictors and the dependent variable, meaning it may fail to capture nonlinear trends or threshold effects. It is also highly sensitive to outliers and influential data points, which can distort model estimates and lead to misleading conclusions. Furthermore, stepwise selection can sometimes lead to overfitting, particularly when many predictors are tested, and underfitting when potentially meaningful variables are excluded. In our data specifically, some variables show skewed distributions or restricted ranges, which could affect coefficient stability. In addition, verifying that our model satisfied the assumptions of multiple linear regression was important, for unmet assumptions could lead to greater limitations in the analysis.

Regarding the linearity assumption, we assessed this concern utilizing a fitted-versus-residual scatterplot. Demonstrated above in the left plot of Figure 4, there is a notable sharp border created by the datapoints near zero in a declining format. This makes sense because our data is constrained to zero or positive numbers, whereas the fitted values are not. Therefore, despite the sharp boundary, the rest of the points do not follow any systematic pattern, suggesting that the relationships in the model are generally linear. Independence of errors was examined using the Durbin-Watson test, which resulted in an insignificant p-value of 0.2539, indicating that residuals were not significantly autocorrelated and supporting the assumption of independence. Regarding the assumption of constant error variance, residual plots revealed strong heteroscedasticity, suggesting that the spread of residuals increases with fitted values. This may indicate model misspecification or unaccounted variability, meaning that the assumption of constant error variance was not met. For example, if the variance of residuals increases with the number of people invited, the model is less certain about predictions for high values. Next, the normality of residuals assumption was satisfied, as shown by the histogram on the right in Figure 4,

which displays an approximately normal distribution. Lastly, multicollinearity diagnostics revealed that all of our predictor variables had low VIF values, confirming that this assumption has also been met.

Based on our model results, several variables appear to have strong and statistically significant relationships with event attendance. RSVP behavior is one of the most influential predictors, with each additional “Yes” RSVP being associated with an estimated increase of about 0.45 attendees and each “Maybe” RSVP being associated with roughly 3.82 fewer attendees on average, likely reflecting overestimation of attendance among preliminary respondents. In practice, this suggests that hosts should treat “Yes” RSVPs as meaningful indicators of turnout, while “Maybe” RSVPs are negatively associated with turnout and may indicate that fewer attendees will ultimately arrive.

The number of approved individual invitations (``numApprovedID``) shows a strong positive effect, with an estimated coefficient of 0.50. This suggests that, on average, that about half of those who are invited to a given event actually attend. However, the interaction between the number of organizations invited (``numAllowedOrgs``) and the number of individuals invited is negative and significant, suggesting diminishing returns: as more organizations are invited, each additional approved invitation contributes less to overall attendance. This suggests that inviting additional organizations does not necessarily increase turnout in proportion to the number of people invited. Due to the approved ID list already includes members of all invited organizations, adding more organizations may broaden the invitation pool without meaningfully increasing the number of attendees who actually show up. In practice, hosts may experience reduced engagement per invited person as the number of invited organizations grows.

Timing also has a moderate but meaningful impact, such that events held later in the day (``event_hour``) tend to attract slightly higher attendance, consistent with expectations for many of the social activities in our dataset. Weekend events show somewhat lower average attendance than weekday events, which is to be expected since weekday events are often academic-based, whereas weekend events tend to be more leisurely. However, this difference is not statistically significant, suggesting that event timing may be less influential than other characteristics.

Regarding event type, certain categories (e.g., “dFunc” events) show significantly higher attendance, while others such as mixers or meetings draw smaller crowds. These differences highlight the value of understanding how event purpose influences turnout and can inform targeted planning for different event styles. Event planners can leverage these patterns by anticipating higher turnout for certain event types and managing resources accordingly.

Overall, the model explains roughly 65% of the variation in attendance (Adjusted $R^2 = 0.64$), suggesting that the included variables capture much of the systematic pattern in event participation. This indicates that the model can meaningfully support event planning decisions by providing hosts with evidence-based expectations of turnout. With the combination of statistical outputs and diagnostic checks, we have ensured that our final model is both interpretable and reasonably well-suited for making predictions based on the data.

Dataset Overview

The dataset contains **1,213 UVA events** hosted through the DoorList app from **August 5th, 2024 to September 8th, 2025**. These events span a wide range of categories—big parties, meetings, mixers, date functions, small gatherings, and more. Each row includes details such as:

- Event name
- Hosting organization
- Start time and day
- Visibility settings (public vs. restricted)
- Whether the event was cancelled
- Attendance-related information (scanned attendees, approved IDs, RSVPs, invited organizations, plus-ones)

These variables gave us insight into how DoorList is used across Grounds and what factors might be associated with event turnout.

Initial Data Cleaning and Filtering

The raw data was provided in **Google Sheets** and stored in **JSON format**, which required careful parsing to extract usable information. Before analysis, we removed two categories of events from the dataset:

1. **Test cases** provided by DoorList representatives
2. **Cancelled events**, filtered by removing rows where `eventCancelled == TRUE`

Because many events in the raw dataset had **0 attendees**, we also applied a validity filter. We kept:

- All events with **at least one invited organization**, even if attendance was 0 (since legitimate events sometimes have no scans)
- Events **without invited organizations** only if they had **10 or more attendees**

This allowed us to remove “joke” or inactive events while keeping legitimate but unscanned ones. After filtering, the dataset was reduced to **635 events**.

Parsing and Creating Variables

To prepare the data for modeling, we parsed several JSON-based columns and created new variables:

- **attendee_count**: number of IDs in `thruDoor`
- **numApprovedID**: count of IDs in `approvedID`
- **totRequestedPlusOnes**: count of plus-one requests

- **RSVPyes, RSVPmaybe, RSVPnotGoing**: counts of RSVPs
- **numAllowedOrgs**: number of organizations invited
- **orgPlusOnes_num**: number of organizational plus-ones

We also converted event timestamps from **UTC to Eastern Time** and extracted:

- **event_hour** (0–23)
- **event_day** (day of week)

These new time-based variables helped us identify when events were most commonly held and whether certain days or times correlated with attendance.

Feature Engineering: Creating “event_type”

Based on our knowledge of UVA social life and DoorList usage, we created a new variable, **event_type**, to categorize the type of event. Since our cleaned dataset contained 635 events and our team has four members, we divided the classification work among us.

We ultimately established **eight categories**:

1. Big party
2. Mixer (2–4 organizations)
3. Me + 3
4. Rush event
5. Meeting
6. Date function
7. Small event
8. Other

To classify events consistently, we created a **data dictionary** linking keywords from event names to categories—for example, “dfunc,” “date function,” and “df” mapped to the date function category.

Forming Our Research Question

Once we understood the structure of the data, we began considering what research question we could answer meaningfully with event-level information. Given that DoorList is designed to help hosts manage attendance, we chose to focus on a relevant and practical question:

How many people actually attend an organization's events?

This question fit the purpose of the app and would help hosts better plan event logistics. By analyzing variables such as event type, timing, invited organizations, RSVP patterns, and plus-one requests, we aimed to identify which characteristics most strongly relate to event turnout.

Why the Data Was Useful—And Its Limitations

Although the DoorList dataset provided rich event-level information, several limitations prevented us from exploring more nuanced questions:

- **Lack of person-level data**
This prevented us from studying individual engagement, friend-group attendance patterns, or student-level behavior over time.
- **Only one year of data**
Multiple years would have allowed us to examine long-term trends, recurring event attendance stability, or semester-based differences.
- **Missing external factors**
Additional information such as:
 - Weather conditions
 - Competing non-DoorList events
 - Methods of event promotion
- would have given us more insight into what drives attendance beyond the variables available.

Despite these limitations, the event-level dataset provided a valuable opportunity to analyze how UVA organizations host events and what factors influence turnout. Working with raw JSON-based data also gave us meaningful experience with a complex, real-world dataset.

After obtaining the dataset from DoorList, we started by analyzing it to identify possible information we could utilize for our project. The data is at the event level, meaning each row represents a single event rather than individual attendees, which allows us to focus on how events differ from one another. . Although the data is at the event level rather than the individual level we initially sought, and stored in JSON format, which requires parsing each cell to accurately extract values, we saw it as an opportunity to work with a complex dataset. This way, we could answer a research question that is meaningful to us along with other UVA students.

The raw data source was given to us as a Google Sheets document containing UVA event-level DoorList data from the past year, specifically from August 5th, 2024 to September 8th, 2025. The data contained 1213 rows of UVA events, including parties, meetings, date functions, and more. There were 117 columns in the dataset, including many predictor variables and our response variable, thruDoor, which lists the IDs of people who attended an event via DoorList's scanning feature. After discussions

with DoorList representatives and internally with our team, we concluded that including all 116 predictor variables in our model would not be practical. We made this decision for several reasons: many predictors are irrelevant to our analysis, some contain mostly missing values, others are dated and no longer in use, and, most importantly, having too many predictors would hinder our model's performance. So, after reading in the raw csv file to our R markdown document, we decided to filter the dataset to only include 13 of those original 116 predictors; we felt that these 13 variables were most relevant to our analysis. These variables are name (name of event), allowedOrgs (IDs of organizations invited to event), eventCancelled (boolean for if event was cancelled), orgFBid (ID of organization hosting event), time (start time), orgName (name of organization hosting event), approvedID (IDs of people on the guest list), requestedPlusOnes (ID and number of plus ones they requested), visibility (if event is public or private), orgPlusOnes (organization ID and how many plus ones each person in that organization gets), going (IDs of people who RSVP "Yes"), maybe (IDs of people who RSVP "Maybe"), and notGoing (IDs of people who RSVP "No"). Additionally, we filtered the dataset further by removing Doorlist's test cases and events that were cancelled. DoorList representatives gave us a list of organization names that were associated with test cases, and we removed these events with these organization names from our dataset. To determine which events were cancelled, we just removed events with TRUE in the eventCancelled field.

Next, in order to better format this data for our analysis, we had to parse through fields and make additional columns with these parsed values, since the raw format was JSON. First, we parsed through the thruDoor column, our response variable, by counting the number of commas separating the different attendee IDs and adding 1 to accurately represent the total number of attendees. If there was an empty list of IDs, the attendee count for that event was set to 0. After parsing thruDoor, we created a new column, attendee_count, which counts the number of attendees at each event. We followed the same approach for other columns like approvedID, which counts the number of approved IDs for an event and totals them in a column named numApprovedID. For other columns that we needed to parse through, such as requestedPlusOnes, going, maybe, notGoing, allowedOrgs, and orgPlusOnes, we followed similar procedures to count the IDs and made columns such as totRequestedPlusOnes, RSVPyes, RSVPmaybe, RSVPnotGoing, numAllowedOrgs, and orgPlusOnes_num respectively.

Finally, we also changed the time variable from UTC format to Eastern time, and extracted both the hour and day of the week from each timestamp to make the columns event_hour and event_day. Event_hour is represented by a number from 0-23, and event_day is represented by the day of the week. With these new variables, we were able to make our time-based analyses more interpretable and uncover patterns in event activity throughout the day and week.

During our initial exploratory data analysis, we observed that many events in the dataset had attendee counts of 0. To address this, we decided to focus on a subset of events: those with at least one invited organization, regardless of attendance size, and those without invited organizations only if they had 10 or more attendees. We made this choice because many events appeared to be "joke" events created by users that never actually took place. However, we still included events with 0 attendees if at least 1 organization had been invited, since some legitimate events are listed on DoorList but have no scans, which is a pattern we have observed from experience. This approach allowed us to retain valid but

unscanned events while filtering out those that were likely invalid. After filtering, we were left with 635 events, meaning 635 rows in our dataset.

Based on our understanding of UVA events and DoorList, we decided to add a new variable, `event_type`, to classify each event, as we believed that it could improve our ability to predict the response variable, `attendee_count`. Since our filtered dataset included 635 events and our team consists of 4 of us, we decided to divide the events among ourselves and individually classify each type. We came up with 8 categories to classify each event for the `event_type` variable: big party, mixer (get together with 2-4 fraternities/sororities), me + 3 (fraternity/sorority party where each person in the organization gets to invite 3 guests), rush event, meeting, date function, small event, and other. We relied on our judgement and experience as UVA students to assign each event to the most appropriate category. To classify events in our model, we created a data dictionary that maps specific keywords from event names to their corresponding categories, for example, for date function some key words are `dfunc`, `date function`, `df`, etc.

After familiarizing ourselves with the data and gaining a better sense of what variables we had, we started thinking about how we could use it to learn more about event attendance at UVA, and hopefully generalize the information to more schools, although this could be difficult. Since our data is structured at the event level, we decided to focus on attendance and understand what factors might influence how many people show up to an event. The main research question we chose to study is: *How many people actually attend an organization's events?*

We decided on this question because it closely aligns with how DoorList is actually used. The app is centered around tracking attendance and aiding hosts in organizing who comes to their events, so we thought focusing on turnout fit with DoorList's purpose. Rather than measuring how "successful" an event is, our goal is to predict how many people attend so hosts can plan more effectively. By examining factors such as event type, time and day, number of invited organizations, RSVP responses, among others, we can hopefully identify characteristics closely connected to event turnout. Understanding these patterns can help organizations plan their events more effectively and make smarter decisions about scheduling, capacity, and preparation.

While the dataset we received from DoorList had a lot of data to work with, there were still a few things we wish we could have included. The main limitation was that our data was at the event level instead of an individual, or person, level. Having individual-level data would have allowed us to explore student engagement on grounds in a more specific way. For example, we could have potentially found patterns like whether friends or people in the same organizations tend to go to similar events, if students in certain groups prefer particular event types over others, how people's attendance habits change over time, etc. It would have been interesting to study and include how individual and/or group dynamics contribute to event attendance.

We also wish we could have had access to more than one year of data. With multiple years of data, this would have provided more information about how events and attendance have changed over time and whether any long-term patterns exist. It would have been helpful to look at recurring events to see if attendance stayed consistent or shifted from year to year, or if certain even types became more or

less common over time. With additional years of data, we also could have examined whether hosting activity varies by semester to see if certain parts of the year consistently have higher event activity and attendance than others.

In addition to wanting person-level and multi-year data, there were a few other types of information that would have been beneficial for our analysis but weren't included in the dataset. For example, weather data could have helped explain variation in turnout, since factors like rain, cold temperatures, or warm, sunny days can affect whether people choose to attend an event. It also would have been useful to know if there were other non-DoorList events occurring at the same time, since those could draw attendance away from DoorList events. A final piece of information that could have strengthened our project is how events were promoted, such as whether they were advertised more publicly across grounds or mainly within smaller groups or organizations. This additional information could have given us a clearer understanding of what factors drive attendance and how external conditions might influence the patterns we saw in the DoorList data.